

A Rosenthal-positive unipotent orbit in $\mathrm{SL}_2(\mathbb{R})$

Run `fa.banach_001`, lane 6

June 18, 2026

Status

Partial result. This packet proves that a natural homogeneous subspace G/H of the conjugation G -space G_c , for $G = \mathrm{SL}_2(\mathbb{R})$, is Rosenthal representable. It does not decide whether the whole conjugation action $G \curvearrowright G_c$ is Rosenthal representable.

Source question and context

The source paper is Michael Megrelishvili, *Topological group actions by group automorphisms and Banach representations*, arXiv:2110.01386. The main concrete question asks:

in contrast, the left action of S_∞ on itself is always Hilbert representable and the topological group S_∞ is unitarily representable on a Hilbert space.

Question 3.20. *Is it true that for $G = \mathrm{SL}_2(\mathbb{R})$ the G -space G_c is Rosenthal representable?*

4. BANACH REPRESENTATIONS OF HOMOGENEOUS ACTIONS

The same paper observes that the proof of non-reflexive representability for G_c already takes place on the unipotent conjugacy orbit of

$$u = x_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

and identifies this orbit with a locally compact coset space G/H . It then asks which homogeneous G -spaces G/H are Rosenthal representable:

Proof. Indeed, in Theorem 3.3 we deal, in fact, with the restricted action by conjugations of $G := \mathrm{SL}_2(\mathbb{R})$ on the G -orbit of x_1 . This orbit, containing every

$$x_m := \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix}$$

(in terms of Theorem 3.3) is locally compact and it can be represented as the topological coset G -space G/H , where H is the stabilizer subgroup

$$\mathrm{St}(x_1) = \{g \in \mathrm{SL}_2(\mathbb{R}) : gx_1g^{-1} = x_1\}.$$

□

Results of the present section suggest the following general questions.

Question 4.4. *Which interesting homogeneous G -spaces G/H are Rosenthal representable? What about the $\mathrm{SL}_n(\mathbb{R})$ -spaces $\mathrm{SL}_n(\mathbb{R})/H$? In particular, what if $H = \mathrm{SL}_n(\mathbb{Z})$?*

The “Asplund version” of Question 4.4 seems to be less attractive than the present “Rosenthal version”. The reason is that by [92, Theorem 6.10] and Fact 2.5, Asplund

Proof Intuition

The unipotent orbit is not an arbitrary piece of G_c . It is the cone of nonzero rank-one nilpotent matrices in $\mathfrak{sl}_2(\mathbb{R})$, translated by the identity. This cone is parametrized by nonzero vectors $v \in \mathbb{R}^2$, up to the sign $v \sim -v$, through a quadratic map $v \mapsto N_v$.

Megrelishvili proves that the one-point compactification of the linear $\mathrm{GL}_2(\mathbb{R})$ -action on $\mathbb{R}^2$ is tame. The key observation here is that the one-point compactification of the unipotent orbit is a $\mathrm{SL}_2(\mathbb{R})$ -factor of that tame compactification: both the zero vector and the point at infinity are sent to the single added point of the unipotent orbit compactification. Tameness passes to factors, hence the orbit is Rosenthal representable.

The theorem

Theorem 1. *Let $G = \mathrm{SL}_2(\mathbb{R})$, let*

$$u = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

and let

$$H = \mathrm{St}(u) = \{g \in G : gug^{-1} = u\}.$$

Then the homogeneous G -space G/H , identified with the conjugacy orbit $G \cdot u \subset G_c$, is Rosenthal representable.

Proof. Let $\omega(v, w) = \det(v, w)$ be the standard symplectic form on $\mathbb{R}^2$. For $v \in \mathbb{R}^2$, define a linear operator

$$N_v(w) = \omega(v, w)v.$$

In coordinates $v = (a, b)$,

$$N_v = \begin{pmatrix} -ab & a^2 \\ -b^2 & ab \end{pmatrix}.$$

Thus $N_v \in \mathfrak{sl}_2(\mathbb{R})$, $N_v^2 = 0$, and $N_v \neq 0$ for $v \neq 0$. Also $N_{-v} = N_v$, and $N_{\lambda v} = \lambda^2 N_v$.

For every $g \in G$, preservation of ω gives

$$gN_v g^{-1} = N_{gv}.$$

Since every nonzero vector of $\mathbb{R}^2$ is the first column of some matrix in $\mathrm{SL}_2(\mathbb{R})$, the conjugacy orbit of $u = I + N_{(1,0)}$ is exactly

$$O := G \cdot u = \{I + N_v : v \in \mathbb{R}^2 \setminus \{0\}\}.$$

This orbit is G -equivariantly homeomorphic to G/H .

Let $X = \mathbb{R}^2 \cup \{\infty_X\}$ be the Alexandrov one-point compactification of $\mathbb{R}^2$, equipped with the natural G -action. Let $Y = O \cup \{\infty_O\}$ be the Alexandrov one-point compactification of the locally compact orbit O . Define

$$q : X \longrightarrow Y$$

by

$$q(v) = I + N_v \quad (v \neq 0), \quad q(0) = q(\infty_X) = \infty_O.$$

The map is onto and G -equivariant by the identity $gN_v g^{-1} = N_{gv}$.

It remains only to check continuity at the two points collapsed to ∞_O . Let $K \subset O$ be compact. Since O is embedded in the Hausdorff group G , K is closed in G . If $v_i \rightarrow 0$ and $I + N_{v_i} \in K$, then $I + N_{v_i} \rightarrow I$ in G , forcing $I \in K$, impossible because $I \notin O$. Hence vectors sufficiently close to 0 are mapped outside K . Similarly, if $\|v_i\| \rightarrow \infty$, then the matrix entries of N_{v_i} are unbounded, so $I + N_{v_i}$ eventually leaves every compact subset of O . These are precisely the neighborhood conditions for continuity at 0 and at ∞_X . Thus q is a continuous G -factor map from X onto Y .

By Theorem 3.12 of [1], the linear $\mathrm{GL}_2(\mathbb{R})$ -action on $\mathbb{R}^2$ has tame one-point compactification. Restricting the acting group from $\mathrm{GL}_2(\mathbb{R})$ to $G = \mathrm{SL}_2(\mathbb{R})$ preserves tameness, and tameness passes to compact factors. Therefore Y is a tame compact metric G -space. By Fact 3.11 of [1], compact metric tame systems are exactly the systems representable on Rosenthal Banach spaces. Since Y is a proper G -compactification of $O \simeq G/H$, the homogeneous G -space G/H is Rosenthal representable. $\square$

Corollary 1. *For $G = \mathrm{SL}_2(\mathbb{R})$, the stabilizer $H = \mathrm{St}(u)$ gives a homogeneous space G/H which is Rosenthal representable but not reflexively representable.*

Proof. Rosenthal representability is the theorem above. Non-reflexive representability is exactly the source paper's Corollary 4.3, whose proof says that Theorem 3.3 works on this restricted conjugacy orbit. $\square$

Limitations

This does not settle Question 3.20. Tameness of a G -subspace or a single orbit compactification does not imply tameness of the whole conjugation G -space G_c . The result only shows that the specific unipotent orbit responsible for the paper's double-limit non-WAP argument cannot also serve as a non-Rosenthal witness.

Novelty and literature check

Local run indexes were searched for arXiv id 2110.01386, the title, $\mathrm{SL}_2(\mathbb{R})$, Rosenthal representability, unipotent orbit, and the stabilizer notation $\mathrm{St}(x_1)$; no duplicate packet or attempt was

found. Web searches on June 18, 2026 used exact combinations of $\mathrm{SL}_2(\mathbb{R})$, Rosenthal representability, $St(x_1)$, unipotent orbit, and Megrelishvili. They found the source paper but no later resolution or prior statement of this factor-map consequence.

References

- [1] M. Megrelishvili, *Topological group actions by group automorphisms and Banach representations*, Forum Mathematicum, accepted version; arXiv:2110.01386, 2023.